

Andrey Derzhavin

+7 921 389-90-39 | derzhavin@frtk.ru | github.com/derzhavin3016 | linkedin.com/in/andrey-derzhavin/ | t.me/derzhav1n

SUMMARY

CPU simulation engineer with 5+ years at Huawei. I led a functional CPU simulator from prototype to product integration, designed a parallel framework for microarchitectural exploration, and built loop-analysis tooling on top of Boost. Core stack: C++, JIT compilation, AArch64 assembly, LLVM. Co-created and taught the “CPU & OS Simulation” university course. Looking for roles in emulation, simulation, or high-performance systems.

EXPERIENCE

Huawei Jul 2021 — Present
Software Engineer *Moscow, Russia*

- Co-developed and co-taught “CPU & OS Simulation” course (MIPT, ITMO)
- Led development of a functional CPU simulator from early prototype to production integration within one year (team of two); owned architecture, implementation, and delivery.
- Researched microarchitectural design space exploration using low-order parameters. Designed a parallel, simulator-agnostic framework that identifies optimal microarchitectural configurations.
- Built a static loop analysis tool using Boost Graph Library. Studied dynamic loop classification and quantified potential IPC improvement from enhanced loop termination prediction via ISA hints.

Acronis Jul 2020 — Nov 2020
Intern *Remote*

- Investigated right-context grammars for static analysis of C++ source code; proposed several novel applications.

EDUCATION

Moscow Institute of Physics and Technology Dolgoprudny, Russia
M.Sc. in applied math and physics, honours degree Sep 2023 — Jul 2025

Moscow Institute of Physics and Technology Dolgoprudny, Russia
B.Sc. in applied math and physics, honours degree Sep 2019 — Jul 2023

- Selected Coursework:** “Uses and Applications of C++ Language” (MIPT, 2020–2021), lecturer K. Vladimirov (Intel).

Physics and math lyceum 30 Saint-Petersburg, Russia
Sep 2017 — Jun 2019

PROJECTS

leech-compiler (github.com/derzhavin3016/leech-compiler) Sep 2023 — May 2024

- Designed an SSA-based compiler IR for a custom VM. Implemented optimization passes (constant folding, inlining, peepholes, checks elimination).

JitResearch (github.com/ProteusLab/JitResearch) Sep 2025 — Present

- Comparative study of modern JIT frameworks (LLVM ORC, AsmJit and other JIT backends) for CPU functional simulation. Evaluated compilation latency, code quality, and integration complexity. Results presented at C++Russia 2026.

PUBLICATIONS & TALKS

JIT libraries for CPU simulation: difficulties of choice (cppconf.ru/talks/20010048/) May 2026

- Talk at C++Russia 2026 (accepted). Discussed trade-offs among JIT libraries for functional CPU simulation.

TECHNICAL SKILLS

- Programming & Assembly:** C, C++, Python, AArch64, RISC-V, x86-64
- Simulation & Emulation:** QEMU, gem5, functional CPU simulators
- Compilers & IR:** LLVM IR, JIT compilation
- Libraries & Frameworks:** Boost, Google Test, OpenCL, NumPy
- Tools & Platforms:** git, make, CMake, Linux (WSL2, Ubuntu)
- Languages:** Russian (native), English (C1, IELTS 7.0)